

Backend engineer building production AI systems for AI/3D products, specializing in distributed workers, async pipelines, and agent orchestration. Improved service stability and contributed to business growth through global AI launch, reusable operating models for long-running jobs, and 3D automation pipelines — all under tight resource constraints, with multiple internal engineering and AI awards for 3D automation and AI advancement. Turned async worker patterns, autoscaling, and failure isolation into reusable building blocks shared across multiple services. When needed, filled FE/ML/infrastructure gaps to keep delivery moving, but always anchored in backend operating models and domain design. Recently designing and building an AI Agent Platform that orchestrates diverse interior agents.

Experience

Bucketplace

📅 Aug 2022 – Present

Senior Backend Engineer

- **Ohouse AI Launch:** Completed global launch to **170 countries in 4 weeks**, accelerating a 6-week target. Built queue-based async workers with autoscaling, failure isolation, and reusable retry/state patterns to separate long-running AI jobs from user-facing APIs and handle unpredictable traffic safely.
- **Ohouse AI Monetization / Evolution:** Built the iOS/Google subscription system from scratch with a unified IAP state machine, designed RAG-based affiliate recommendations achieving **CVR 10.5%**, implemented provider failover, and migrated to **Python/Langgraph** for extensible AI workflows.
- **Panorama AR:** Led the backend for a service that reconstructs users' rooms in 3D for virtual furniture placement. Designed and launched microservices to productionize ML research results, with async state management for long-running 3D Reconstruction pipelines, handling **~300 daily room scans**.
- **Room Planner:** Improved performance and reliability, contributing to **704% WAU growth** and **600% annual GMV (KRW 860M)** while sustaining **99.95% availability** at **10.2x traffic**.
 - API latency **200ms→80ms**, payload **54%** reduction, **LoD auto-generation** for **90%** model size reduction to make heavy 3D scenes viable on mobile. Draw calls **3,000→20 (99%)**, model loading **10s→2.5s**, resolving mobile rendering bottlenecks.
 - Load-tested event/challenge spike scenarios for **zero-incident** operation. Expanded AR-supported products from **2,210 to 231,000 (104x)** via AR Size Box, lifting purchase conversion by **1.7x**.
 - Redesign associated-item recommendations from real-time to an **offline scoring algorithm**, cutting search API latency by **99.7%**.
- **3D Asset Automation:** Automated the manual 3D production pipeline, scaling monthly output from **68 to 800 items** and reducing cost by **88%**, earning the **Eng Award**.
 - Built a Digital Asset Management (DAM) system — GPU **20%** reduction, file size **51%** smaller, **40+ days/year** saved (**Top Contributor**). Completed E2E automation with GPT-4o-based quality scoring across 12 criteria.
- **Technical Leadership:** Onboarded new hires, FE, and ML collaborators via **code reviews** so they could contribute to the backend independently.
 - Shortened release cycles with release-note and analytics automation, earning **AI Award** and **AI Native MVP** recognition.

LG CNS

📅 Jan 2020 – Jul 2022

System & Solution Programming Consultant

- Designed and operated a Node.js + Redis real-time sync server, building a WebSocket-based multi-user concurrent editing environment.
- Automated CI/CD pipelines with GitLab/Jenkins to reduce build, test, and deployment lead time, and developed backend APIs for Unity client integration.

Education

University of Seoul | Master's in Computer Science |

📅 2018 – 2019

Research on improving AI system performance / Early completion of Master's degree through a combined Bachelor's-Master's program.

University of Seoul | Bachelor's in Computer Science

📅 2012 – 2017

Ohouse AI: Launch and Evolution

Problem

An AI service that auto-places furniture in real interior photos needed to launch globally within 6 weeks with limited staffing and many cross-team dependencies.

Multi-turn image generation required strict per-user ordering and quota-limit handling, and external AI failures had to be contained.

Approach

Separated API and workers via message queues, then ran Kubernetes autoscaling with concurrency limits for independent scaling.

Designed retry/isolation policies and unblocked the launch with JWT-based guest auth while the auth server was still in progress.

Post-launch, applied Apple/Google IAP state machine, LLM Provider Fallback + Circuit Breaker with AI provider failover, and migrated to Python/Langgraph graph-based architecture for monetization and scalability.

Designed RAG-based affiliate recommendation path under Amazon Affiliate policy constraints for real-time product suggestions.

Result

Completed launch to **170 countries in 4 weeks** and maintained uninterrupted operation under unpredictable traffic with autoscaling and backpressure.

Recovered quickly from external failures via provider failover architecture.

Achieved affiliate **CVR 10.5%**, proving the monetization model.

3D Asset Pipeline & Production Automation

Problem

3D models for Room Planner and AR features were produced manually.

Existing assets lacked a management system, wasting GPU and storage, while new model creation couldn't scale due to throughput limits and quality variance.

Approach

Built a Digital Asset Management (DAM) system for deduplication, auto-optimization, and deployment pipeline, with an ops dashboard integrating 3D preview and AR/Room Planner enabling non-engineer review.

Automated the Image-to-3D pipeline with state-based workflows and DB-tracked worker progress for recovery and scaling.

Stabilized quality through parameter tuning and fallback logic, then completed E2E automation with GPT-4o-based quality scoring across 12 criteria.

Result

DAM reduced GPU by **20%**, file size by **51%**, and saved **40+ days/year**, earning **Top Contributor**.

Image-to-3D automation scaled monthly output from **68 to 800**, cut cost by **88%**, and expanded **45 new categories**, earning the **Eng Award**.

Room Planner Performance and Reliability

Problem

In a mobile 3D interior layout service, slow initial loading was the key bottleneck driving user drop-off and stalling growth.

Event/challenge traffic spikes required zero-downtime operation, and rendering speed was critical for session retention.

Approach

Performance: Lightened 3D models by **90%** via LoD auto-generation and mesh simplification, and cut API latency from **200ms to 80ms** by restructuring API call patterns. Reduced draw calls from **3,000 to 20** and optimized the rendering pipeline, cutting still-image rendering from **60s to 20s** and animation from **20min to 3min**.

Reliability: Linked queue metrics to autoscaling rules with safe shutdown behavior, and validated event/challenge spike scenarios through load testing.

Result

Performance gains increased average session time from **21min to 35min (68%)**, contributing to **704% WAU growth** and **600% annual GMV (KRW 860M)**.

Maintained **zero incidents** during traffic surges while sustaining **99.95% availability** at **10.2x traffic**.